



RESEARCH HIGHLIGHT

Into the Spotlight: Microglia in the Peripheral Nervous System

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Microglia have been recognized as a unique cell population in the central nervous system (CNS) for more than a century [1]. However, it was not until 2010 that their developmental origin was clarified. Rather than arising from the neuroectoderm, microglia are derived from erythromyeloid progenitors in the embryonic yolk sac [2]. Following this paradigm shift, an increasing number of studies began to reveal the heterogeneity, diverse functions, and pathological relevance of microglia. As resident macrophages of the CNS, microglia play critical roles in neurogenesis, synaptic pruning, immune surveillance, and pathogen defense. Traditionally, microglia have been considered exclusive to the CNS, while

only conventional tissue-resident macrophages were thought to populate the peripheral nervous system (PNS) [1, 3].

A recent study by Wu *et al.* [4] published in *Cell*, challenges this long-standing assumption. The authors identified a previously unrecognized population of microglia-like cells in the PNS, expanding our understanding of microglial distribution.

Previously, those authors have identified microglia-like cells in various peripheral organs, including fetal skin, testicles, and hearts in human prenatal samples [5]. In this more recent study, they performed cross-species single-cell RNA sequencing (scRNA-seq) on samples from adult humans, mice, rats, pigs, and crab-eating macaques. By isolating immune cells from both the CNS (brain and spinal cord) and peripheral ganglia (dorsal root ganglia [DRG] and sympathetic ganglia [SG]), they identified a population of cells in the periphery that share signature genes with CNS microglia. These PNS microglia-like cells expressed core microglia markers, including *P2RY12*, *TMEM119*, and *SALL1*, but lacked markers for classical macrophages such as *MRC1*, *DAB2*, and *LYVE1*. Notably, this population was present in larger mammals, including humans, macaques, and pigs, but absent in rodents such as mice and rats. The presence of PNS microglia-like cells was validated using immunofluorescence staining and flow cytometry (*P2RY12*⁺*MRC1*[−]). In addition, Wu *et al.* performed ATAC-seq and ChIP-seq analyses to characterize the epigenetic landscape of PNS microglia-like cells. By integrating chromatin accessibility, histone modification, and binding of the canonical microglia-defining transcription factor PU.1 at marker gene loci, Wu and colleagues demonstrated that PNS microglia-like cells closely resemble CNS microglia while being distinct from conventional macrophages.

Do PNS microglia-like cells share developmental and differentiation trajectories with CNS microglia? To address

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this question, the authors took advantage of their technical strengths by performing temporally resolved scRNA-seq of human embryonic hematopoietic cells from different developmental stages, followed by clustering and trajectory analyses. They found that, similar to CNS microglia, PNS microglia-like cells originate from yolk sac-derived macrophage progenitors and differentiate through a distinct PNS microglia-like cells precursor state. As development progresses, these cells increasingly gain higher expression levels of microglia markers and lose macrophage markers. *In situ* immunostaining further supported the parallel differentiation of PNS microglia-like cells and CNS microglia. At Carnegie stage 12 (CS12), both the spinal cord and DRG contained P2RY12⁺MRC1⁺ cells, representing precursors of CNS microglia and PNS microglia-like cells. As embryonic development progressed, the number of P2RY12⁺MRC1⁺ cells gradually declined, while CNS microglia or PNS microglia-like cells (P2RY12⁺MRC1⁻) increased correspondingly.

Beyond identifying their presence, it is crucial to determine the physiological roles of PNS microglia-like cells within peripheral ganglia, an essential step in defining their biological importance. Using high-resolution immunofluorescence imaging and 3D reconstruction, Wu and collaborators observed that PNS microglia-like cells enwrap neuronal soma within peripheral ganglia. Interestingly, it has long been thought that satellite glial cells (SGCs), which only reside in the peripheral ganglia, form ‘neuron–glial units’ by wrapping around neuron cell bodies [6]. The identification of PNS microglia-like cells challenges this classical view. By co-staining markers for PNS microglia-like cells, SGCs, and neurons, they revealed a novel tri-cellular structure comprising PNS microglia-like cells, neuron soma, and SGCs. Notably, this PNS microglia-like cells–neuron–SGC structure was observed in humans, macaques, and pigs, but not in rodents. At comparable developmental stages, neurons in the DRG enwrapped by PNS microglia-like cells exhibited larger soma, whereas neurons in the SG (with fewer PNS microglia-like cells) and the ganglia in the enteric nervous system (where PNS microglia-like cells are absent) had smaller soma. To determine the role of PNS microglia-like cells in neuronal maturation, the authors depleted microglia both *ex vivo* in human fetal DRG explants and *in vivo* in pigs using PLX5622 (CSF1R inhibitor). In both models, the depletion led to impaired neuronal soma growth, and the *in vivo* study in pigs shows reduced axonal growth and diminished mechanosensitivity, confirming the crucial roles of PNS microglia-like cells in peripheral neuronal development. In contrast, PLX5622 treatment in rodents lacking PNS microglia-like cells did not affect neuronal soma size.

Wu and colleagues further collected DRG samples from 19 additional species and performed single-cell RNA sequencing and *in situ* immunofluorescence analyses.

This cross-species approach enabled them to investigate the general principles underlying the involvement of PNS microglia-like cells in neuronal development across evolution. DRG sensory neurons vary in size across species, a difference previously speculated to correlate with body size [7]. This study uncovers a deeper underlying principle: species with more PNS microglia-like cells tend to have larger neuron somata in DRG. Both direct visual evidence from confocal imaging and quantified phylogenetic comparative analyses across multiple species support this correlation.

Collectively, this new study by Wu *et al.* reveals the presence of PNS microglia-like cells and provides evidence for their crucial role in the maturation of DRG sensory neurons (Fig. 1). These findings suggest the need to revise the dogma that microglia are restricted to the CNS. A recent Forum article further supports this update by proposing the microglial lineage that includes PNS microglia-like cells based on shared developmental and molecular features [8]. Previous studies of brain microglia have shown that, compared to rodents, human microglia exhibit more significant heterogeneity and distinct expression patterns of key genes [9]. This study further highlights the need to consider the limits of rodent-based research, particularly in the context of microglia biology.

Several important questions will require further investigation. First, while PNS microglia-like cells appear essential for sensory neuron maturation, it is not yet clear how this function is carried out, a challenging question partly due to the lack of genetically tractable non-rodent models. Second, the timing and migratory pathway of PNS microglia-like cells colonization also remains to be clarified. Third, molecular mechanisms by which microglia-like cells in the DRG regulate sensory neuron soma size and how PNS microglia-like cells communicate with neurons and SGCs in general are an interesting open question.

It is also interesting to consider the contrast and comparison between PNS microglia-like cells and CNS microglia. As shown by Wu *et al.*, PNS microglia-like cells differ from CNS microglia in several respects. They form a distinct tri-cellular structure with sensory neurons and SGCs, and are localized around the neuron soma rather than dispersed through the brain parenchyma. While both populations show amoeboid and ramified morphologies, PNS microglia-like cells appear more frequently amoeboid, whereas CNS microglia are often ramified. However, their ultrastructural features, dynamic behaviors such as motility, and turnover mechanisms remain unclear.

Functionally, in the brain, microglia contribute to cognitive processes such as learning and memory [10] and participate in various neurodevelopmental and neuropsychiatric diseases [11, 12]. What physiological roles do PNS microglia-like cells play after development? Might they contribute to peripheral nerve disorders such as chronic pain, an

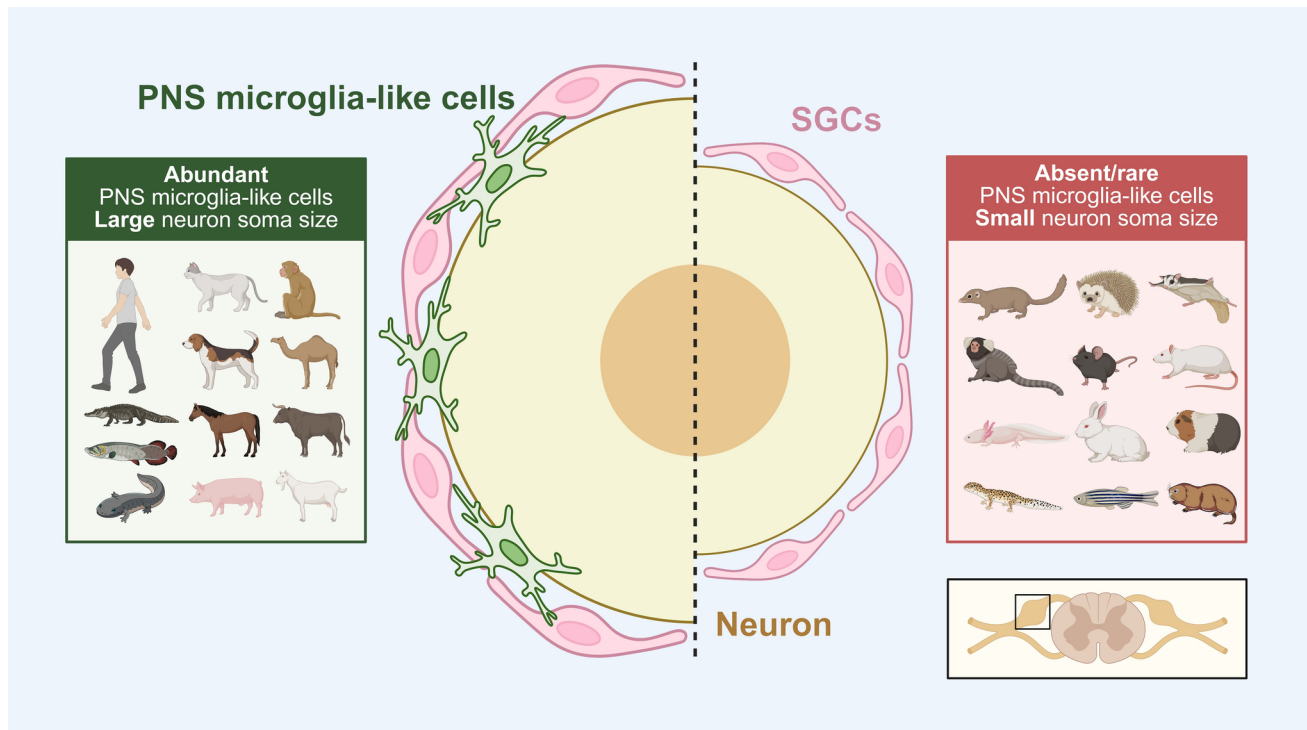


Fig. 1 Schematic of PNS microglia-like cells in sensory neuron enlargement. Species with abundant PNS microglia-like cells, including human, macaque, cat, dog, horse, cow, goat, pig, camel, crocodile, giant salamander, and pirarucu, exhibit larger neuron soma

(left). In contrast, species with rare (e.g., bamboo rat, hedgehog, sugar glider, gecko, axolotl) or absent PNS microglia-like cells (e.g., marmoset, tree shrew, rat, mouse, guinea pig, rabbit, zebrafish) show smaller soma sizes (right). Figure created with BioRender.com.

issue of growing concern? A deeper understanding of PNS microglia-like cells may lead to the identification of novel therapeutic targets and new avenues for the diagnosis and treatment of human diseases.

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Conflict of interest The authors declare that they have no conflict of interest.

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